Tribhuvan University

Institute of Science and Technology

2072

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Bachelor Level / First Year/ First Semester/ Science
Computer Science and Information Technology Stat. 103
(Probability and Statistics)

Full Marks: 60
Pass Marks: 24
Time: 3 hours.

Candidates are required to give their answers in their own words as for as practicable.
All notations have the usual meanings.

Group A

Attempt any Two: (2×10=20)

1. Show that the mean and variance of Poisson distribution are equal. Under which situation Poisson probability distribution would be your good choice?
   If the average number of claims handled daily by an insurance company is 5, what proportion of days have (i) less than 3 claims, (ii) at least 4 claims assuming that the number of claims on different days is independent.

2. Differentiate between confidence level and level of significance.
   A manufacturer of emergency flashlight batteries took a sample of 9 batteries from a day's production and used them continuously until they failed to work. The lifetime (in hours) as measured by the number of hours until failure was:

   33 40 31 54 26 45 70 51 57

   At the 0.05 level of significance, is there evidence that the mean lifetime of the batteries is different from 37 hours?

3. The raw material used in the production of certain synthetic fiber is stored in a location without a humidity control. Measurements of the relative humidity in the storage location and the moisture content of the sample of the raw material were taken 8 days with the following data (in percentage) resulting:

Relative humidity	46	53	29	61	36	39	47	49
Moisture content	12	15	7	17	10	11	11	12

   a) Fit a straight line to the given data by the method of ordinary least squares (OLS) considering moisture content as an outcome variable. Also predict the value of moisture content when relative humidity is equal to 30.

   b) Determine the coefficient of determination and interpret its meaning.

Group B

Answer any eight questions: (8×5=40)

4. If each voter is for proposition A with probability 0.7, what is the probability that exactly 7 of 10 voters are for this proposition? Also compute the probability that at least 3 voters are for this proposition.

5. In the three recent years, the price of copper was 77, 69, 68 cents per pound, and the price of bituminous coal was 19, 18, 20 dollars per short ton. Which of the two sets of prices is relatively more variable?

6. If the probability density function of random variable X is give by:

$f(x) = kx^3$, if $0 < x < 1$

$= 0$, elsewhere

Find the value of k and the probability that the random variable takes on a value: (a) between $\frac{1}{4}$ and $\frac{3}{4}$; (b) greater than 2/3.

7. The burning time of an experimental rocket is a random variable having the normal distribution with mean $\mu = 4.7$ and standard deviation $\sigma = 0.04$ second. What is the probability that this kind of rocket will burn:

(a) less than 4.6 second

(b) more than 4.8 second

(c) anywhere from 4.7 to 4.8 seconds

8 An importer is offered a delivery of laboratory apparatus for \$ 140,000 and the probabilities that he will be able to sell them for \$ 180,000, \$ 170,000 and \$ 150,000 tare 0.3, 0.6 and 0.1. What is the importer's expected gross profit?

9 Obtain the maximum likelihood estimate for the parameter P (proportion of success) of binomial distribution.

10. A sample of 30 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is 0.35 ohms. Determine a 99% confidence interval for the true mean resistance of the population and also interpret your result.

11. Define conditional probabilities. The probability that a circuit chip will have defective drawing is 0.12, the probability that it will have a crack defect is 0.29, and the probability that it has both defects is 0.07. What is the probability that a newly manufactured chip will have either an etching or a crack defects.

12. Prove the following:

(i) $E(aY + b) = aE(Y) + b$

(ii) $E(X_1 X_2) = E(X_1) E(X_2)$.

13. Write short notes on the following:

a) Standard deviation

b) Properties of normal probability distribution